Chapter 9. Building and Extending Loglinear–Logit Models

Collapsibility, ordinal association, rates, sparseness

Contents

9.1 Association graphs and collapsibility **2**
- 9.1.1 Association graphs . . . 2
- 9.1.2 Collapsibility in three-way contingency tables . . . 3
- 9.1.3 Collapsibility and logit models . . . 3
- 9.1.4 Collapsibility and association graphs for multiway tables . . . 3

9.2 Model selection and comparison **4**
- 9.2.1 Considerations in model selection . . . 4
- 9.2.2 Model building for the Dayton student survey . . . 4
- 9.2.3 Loglinear model comparison statistics . . . 5
- 9.2.4 Partitioning chi-squared with model comparisons . . . 6
- 9.2.5 Identical marginal and conditional tests of independence . . . 6

9.3 Diagnostics for checking models **6**
- 9.3.1 Residuals for loglinear models . . . 6
- 9.3.2 Student survey example revisited . . . 7
- 9.3.3 Correspondence between loglinear and logit residuals . . . 7

9.4 Modeling ordinal associations **7**
- 9.4.1 Linear-by-linear association in two-way tables . . . 8
- 9.4.2 Corresponding logit model for adjacent responses . . . 9
- 9.4.3 Likelihood equations and model fitting . . . 9
- 9.4.4 Sex opinions example . . . 9
- 9.4.5 Directed ordinal test of independence . . . 10

9.5 Poisson regression for rates **10**
- 9.5.1 Analyzing rates using loglinear models with offsets . . . 10
- 9.5.2 Modeling death rates for heart valve operations . . . 10

Association graphs and collapsibility (Section 9.1), model selection (Section 9.2), residuals (Section 9.3), ordinal models (Section 9.4) and rate models (Section 9.5).

9.1 Association graphs and collapsibility

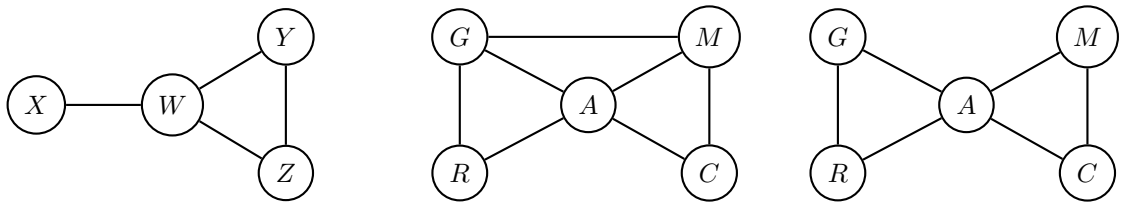
Motivation. Drawing a model as a graph turns conditional independence and collapsibility into questions about paths.

9.1.1 Association graphs

Definition (Association graph, path, separation)

The *association graph* of a loglinear model has a vertex per variable and an edge between two variables iff some term involves both; a *path* is a sequence of edges. A set S *separates* X and Y if every path between them passes through S.

Models with the same pairwise associations have the same graph; e.g. (WX, WY, WZ, YZ) and (WX, WYZ) share Figure 1.



(WX, WY, WZ, YZ) model 6: $(AC, AM, CM, AG, AR, GM, GR)$ model 7: (AC, AM, CM, AG, AR, GR)

Figure 1: Association graphs. Left: book Figure 9.1, W separates X from $\{Y, Z\}$. Middle, right: models for the five-way survey (Example 9.5); in model 7, A separates $\{G, R\}$ from $\{C, M\}$.

Theorem 9.1 (Separation implies conditional independence)

In a hierarchical loglinear model, if a set S of variables separates X and Y in the association graph, then X and Y are conditionally independent given S (after marginalizing over all other variables). More generally, if S separates sets A_0 and B_0, then $A_0 \perp B_0 \mid S$.

Proof. [Supplement]

Step 1. Let V be the set of all variables, A the set of vertices reachable from X by paths avoiding S, and $B = V \setminus (A \cup S)$. Since S separates X and Y, $Y \in B$.

Step 2. Every term T lies in $A \cup S$ or in $B \cup S$: if $T \ni a, b$ with $a \in A$, $b \in B$, hierarchy gives the term $\{a, b\}$, an edge, so $b \in A$, a contradiction.

Step 3. Grouping terms accordingly, $\mu(\mathbf{x}) = f(\mathbf{x}_A, \mathbf{x}_S)\, g(\mathbf{x}_B, \mathbf{x}_S)$ (assign terms inside S to f). Hence, for fixed $\mathbf{x}_S$,

$$P(\mathbf{x}_A, \mathbf{x}_B \mid \mathbf{x}_S) = \frac{f(\mathbf{x}_A, \mathbf{x}_S)\, g(\mathbf{x}_B, \mathbf{x}_S)}{\sum_{\mathbf{a}} f(\mathbf{a}, \mathbf{x}_S) \sum_{\mathbf{b}} g(\mathbf{b}, \mathbf{x}_S)},$$

which factors into a function of $\mathbf{x}_A$ times a function of $\mathbf{x}_B$: $\mathbf{x}_A \perp \mathbf{x}_B \mid \mathbf{x}_S$.

Step 4. Summing over the coordinates of A other than X and the coordinates of B other than Y preserves the product form, so $X \perp Y \mid S$. The statement for sets $A_0 \subseteq A$, $B_0 \subseteq B$ is the same computation. ∎

In Figure 1 (left), W separates X and Y, so $X \perp Y \mid W$ (and also given $\{W, Z\}$); likewise $X \perp Z \mid W$.

9.1.2 Collapsibility in three-way contingency tables

Theorem 9.2 (Collapsibility conditions)

In a three-way table the XY marginal and conditional odds ratios are equal if $Z \perp X \mid Y$ or $Z \perp Y \mid X$, i.e. under (XY, YZ) or (XY, XZ) (or simpler models).

Proof. [Supplement]

Step 1. Under (XY, XZ), by Lemma 8.5, $\mu_{ijk} = A_{ij}C_{ik}$. The conditional odds ratio at level k is

$$\frac{A_{ij}C_{ik}\,A_{i'j'}C_{i'k}}{A_{ij'}C_{ik}\,A_{i'j}C_{i'k}} = \frac{A_{ij}A_{i'j'}}{A_{ij'}A_{i'j}},$$

free of k. The marginal table is $\mu_{ij+} = A_{ij}C_{i+}$, and the row factors $C_{i+}, C_{i'+}$ cancel in the odds ratio, giving the same value.

Step 2. Under (XY, YZ), $\mu_{ijk} = A_{ij}B_{jk}$ and $\mu_{ij+} = A_{ij}B_{j+}$; now column factors cancel, with the same conclusion. ■

Example 9.3 (Collapsibility in the student survey)

In (AM, CM) (graph A–M–C) the AM association is collapsible over C: conditional and marginal fitted odds ratios are both 61.9 (Example 8.8); likewise CM (25.1). AC is not (M is linked to both): conditional 1.0, marginal 2.7. In (AC, AM, CM) nothing is collapsible.

9.1.3 Collapsibility and logit models

Multicenter trial (treatment x, response Y, centers Z): $\operatorname{logit} P(Y = 1 \mid X = i, Z = k) = \alpha + \beta x_i + \beta_k^Z$ is (XY, XZ, YZ), so e^β typically differs from the marginal odds ratio. Without center effects the model is (XY, XZ), collapsible over Z; so with negligible center effects the treatment effect is about the marginal odds ratio.

9.1.4 Collapsibility and association graphs for multiway tables

Theorem 9.4 (Parametric collapsibility, Bishop et al.)

Partition the variables into disjoint A, B, C with B separating A and C. After collapsing over C, the parameters within A and between A and B are unchanged.

Proof. [Supplement]

Step 1. No term meets both A and C (an A–C edge would avoid B). Put terms meeting A into f, the rest into g; terms of f lie in $A \cup B$, those of g in $B \cup C$:

$$\mu(\mathbf{a}, \mathbf{b}, \mathbf{c}) = f(\mathbf{a}, \mathbf{b})\, g(\mathbf{b}, \mathbf{c}), \qquad \log f = \sum_{T \cap A \neq \emptyset} \lambda^T.$$

Step 2. Collapsing over C: $\mu(\mathbf{a}, \mathbf{b}) = f(\mathbf{a}, \mathbf{b})\, G(\mathbf{b})$ with $G(\mathbf{b}) = \sum_{\mathbf{c}} g(\mathbf{b}, \mathbf{c})$. Thus

$$\log \mu(\mathbf{a}, \mathbf{b}) = \sum_{T \cap A \neq \emptyset} \lambda^T + \log G(\mathbf{b}).$$

Step 3. $\log G(\mathbf{b})$ involves only B. By uniqueness of the decomposition of $\log \mu(\mathbf{a}, \mathbf{b})$ under fixed constraints, every term with a variable of A in the collapsed table equals the corresponding λ^T. ■

For (WX, WY, WZ, YZ): $A = \{X\}$, $B = \{W\}$, $C = \{Y, Z\}$ keeps WX after collapsing over Y, Z; $A = \{Y, Z\}$, $B = \{W\}$, $C = \{X\}$ keeps the W, Y, Z associations after collapsing over X. A variable independent of all others can be collapsed over freely. With several variables in B the parameters are unchanged but ML estimates may differ slightly; they agree if the model contains the highest-order term among the variables of B.

9.2 Model selection and comparison

Motivation. Thousands of hierarchical models exist for five variables; inclusion principles and LR comparisons find a simple one that fits.

9.2.1 Considerations in model selection

A model should be complex enough to fit well yet simple to interpret, smoothing rather than overfitting. Guidelines:

(i) Distinguish responses from explanatory variables and concentrate on terms linking responses and linking explanatory variables to responses.

(ii) Include the full interaction among the explanatory variables and any design-fixed margin, so the fit reproduces these totals. For Table 8.8 (responses I, S, predictors G, L) the model should contain GL (e.g. 20,629 urban women, as observed): at least (GL, S, I).

(iii) With a single categorical response, use the equivalent logit model.

(iv) To explore, fit all single-factor, then all two-factor, then all three-factor terms, and so on; use automatic searches with care.

9.2.2 Model building for the Dayton student survey

Example 9.5 (Five-way student survey)

The survey of Example 8.8 also recorded gender (G) and race (R):

		Race = white				Race = other			
		Female		Male		Female		Male	
A	C	M=yes	no	yes	no	yes	no	yes	no
yes	yes	405	268	453	228	23	23	30	19
yes	no	13	218	28	201	2	19	1	18
no	yes	1	17	1	17	0	1	1	8
no	no	1	117	1	133	0	12	0	17

Treating A, C, M as responses and G, R as explanatory, every model contains GR.

Model	G^2	df
1. Mutual independence $+GR$	1325.1	25
2. Homogeneous association (all two-factor terms)	15.3	16
3. All three-factor terms	5.3	6
4g. (2) $-CR$	15.8	17
5. $(AC, AM, CM, AG, AR, GM, GR, MR)$	16.7	18
6. $(AC, AM, CM, AG, AR, GM, GR)$	19.9	19
7. (AC, AM, CM, AG, AR, GR)	28.8	20

Model 3 improves on model 2 by only $15.3 - 5.3 = 10.0$ on 10 df: drop three-factor terms. Backward elimination from model 2 (models 4a–4i; e.g. removing AC gives 201.2, CM 513.5) removes CR (+0.5, df 1), CG (+0.9), MR (+3.2); removing AG would add 5.3 ($P = 0.02$): stop at model 6. (The P-values are not literal, and many counts are small, but G^2 differences remain useful.)

In model 6 (Figure 1, middle) $\{A, M\}$ separates C from $\{G, R\}$: given alcohol and marijuana, cigarette use is independent of gender and race, and by Theorem 9.4 the AC, CM associations equal those of (AC, AM, CM) on the ACM table. Model 7 drops GM, so A separates $\{G, R\}$ from $\{C, M\}$ and all A, C, M associations can be read from the collapsed table. Model 7 fits reasonably ($G^2 = 28.8$, df $= 20$, $\hat{\Delta} = 0.036$).

9.2.3 Loglinear model comparison statistics

Let M_0 be a special case of M_1, with fitted values $\hat{\boldsymbol{\mu}}_0, \hat{\boldsymbol{\mu}}_1$.

Theorem 9.6 (Partitioning G^2)

$$G^2(M_0) = G^2(M_1) + G^2(M_0 \mid M_1), \tag{9.1}$$

where the likelihood-ratio statistic for M_0 against M_1 has the form

$$G^2(M_0 \mid M_1) = G^2(M_0) - G^2(M_1) = 2\sum_i \hat{\mu}_{1i} \log \frac{\hat{\mu}_{1i}}{\hat{\mu}_{0i}}, \tag{9.2}$$

which is the G^2 statistic for M_0 with $\hat{\boldsymbol{\mu}}_1$ playing the role of the data.

Proof.

Step 1. $G^2(M) = 2\sum n_i \log(n_i/\hat{\mu}_i)$, so

$$G^2(M_0) - G^2(M_1) = 2\sum_i n_i(\log \hat{\mu}_{1i} - \log \hat{\mu}_{0i}).$$

Step 2. Write $\log \hat{\boldsymbol{\mu}}_1 = \mathbf{X}_1\hat{\boldsymbol{\beta}}_1$ and, since $M_0 \subseteq M_1$, $\log \hat{\boldsymbol{\mu}}_0 = \mathbf{X}_0\hat{\boldsymbol{\beta}}_0 = \mathbf{X}_1\hat{\boldsymbol{\beta}}_0^*$, where $\hat{\boldsymbol{\beta}}_0^*$ is $\hat{\boldsymbol{\beta}}_0$ with zeros appended for the extra parameters. Then

$$2\mathbf{n}'(\log \hat{\boldsymbol{\mu}}_1 - \log \hat{\boldsymbol{\mu}}_0) = 2\,\mathbf{n}'\mathbf{X}_1(\hat{\boldsymbol{\beta}}_1 - \hat{\boldsymbol{\beta}}_0^*) = 2\,\hat{\boldsymbol{\mu}}_1'\mathbf{X}_1(\hat{\boldsymbol{\beta}}_1 - \hat{\boldsymbol{\beta}}_0^*),$$

using the likelihood equations $\mathbf{X}_1'\mathbf{n} = \mathbf{X}_1'\hat{\boldsymbol{\mu}}_1$ for M_1 (Theorem 8.18). This is $2\sum \hat{\mu}_{1i}\log(\hat{\mu}_{1i}/\hat{\mu}_{0i})$.

Step 3. $G^2(M_0 \mid M_1)$ has the G^2 form. (Note also $\sum \hat{\mu}_{0i} = \sum \hat{\mu}_{1i} = n$ when both models contain an intercept.) Equation (9.1) is the definition rearranged. ■

$G^2(M_0 \mid M_1)$ depends on the data only through $\hat{\boldsymbol{\mu}}_1$. The Pearson difference $X^2(M_0) - X^2(M_1)$ can

be negative; the natural analogue is

$$X^2(M_0 \mid M_1) = \sum_i \frac{(\hat{\mu}_{1i} - \hat{\mu}_{0i})^2}{\hat{\mu}_{0i}}. \tag{9.3}$$

Under M_0 both are asymptotically χ^2 with $\mathrm{df} = \mathrm{df}(M_0) - \mathrm{df}(M_1)$, similar even in fairly sparse tables (Haberman). Under M_1 but not M_0, $G^2(M_1)$ stays χ^2 while the others diverge.

9.2.4 Partitioning chi-squared with model comparisons

Applying (9.1) along nested $M_1 \supset M_2 \supset \cdots$ partitions G^2 into asymptotically independent components. For a $2 \times J$ table let M_j: columns $1, \ldots, j$ form an independence table. Then

$$G^2(M_J) = G^2(M_2) + G^2(M_3 \mid M_2) + \cdots + G^2(M_J \mid M_{J-1}),$$

and $G^2(M_j \mid M_{j-1})$ is the independence G^2 for the 2×2 table of combined columns $1, \ldots, j-1$ versus column j, justifying Section 3.3.3.

For s planned nested tests, level $1 - (1-\alpha)^{1/s}$ each keeps the overall asymptotic type I error at most α; for $s = 3$, $\alpha = 0.05$: 0.017.

9.2.5 Identical marginal and conditional tests of independence

If two direct models differ by one pairwise term, $G^2(M_0 \mid M_1)$ equals the independence G^2 in that pair's marginal table (Sundberg, 1975).

Proposition 9.7 (A conditional test that equals a marginal test)

$G^2[(X, Y, Z) \mid (XY, Z)]$ equals $G^2(X, Y)$ for testing independence in the marginal XY table.

Proof. By Theorem 8.20, $\hat{\mu}_{1,ijk} = n_{ij+}n_{++k}/n$ and $\hat{\mu}_{0,ijk} = n_{i++}n_{+j+}n_{++k}/n^2$, so $\hat{\mu}_1/\hat{\mu}_0 = n\, n_{ij+}/(n_{i++}n_{+j+})$ does not depend on k. By (9.2),

$$G^2 = 2\sum_{i,j,k} \frac{n_{ij+}n_{++k}}{n} \log \frac{n\, n_{ij+}}{n_{i++}n_{+j+}} = 2\sum_{i,j} n_{ij+} \log \frac{n_{ij+}}{n_{i++}n_{+j+}/n},$$

after summing n_{++k}/n over k. This is G^2 for independence in the $\{n_{ij+}\}$ table. ■

This is expected: under (XY, Z) the XY association is collapsible over Z.

9.3 Diagnostics for checking models

Motivation. Residuals show cell by cell where a model fails.

9.3.1 Residuals for loglinear models

Definition (Pearson and standardized residuals)

For cell i the *Pearson residual* is

$$e_i = \frac{n_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i}}, \qquad \sum_i e_i^2 = X^2. \tag{9.4}$$

The *standardized Pearson residual* is $r_i = e_i/\sqrt{1-\hat{h}_i}$, where $\hat{h}_i$ is the leverage (diagonal of the estimated hat matrix).

Pearson residuals have asymptotic variances below 1 (on average residual df/cells); r_i is asymptotically $N(0,1)$ and preferable. Deviance residuals are an alternative.

9.3.2 Student survey example revisited

Five-way survey: with all two-factor terms the only large standardized residual is 3.2 (count 8, fitted 3.1); for model 6, 3.3 (fitted 2.9) in the same cell (nonwhite males smoking cigarettes but using neither alcohol nor marijuana). With so many cells and large n, neither model looks problematic.

9.3.3 Correspondence between loglinear and logit residuals

Proposition 9.8 (Logit versus loglinear Pearson residuals)

For equivalent logit and loglinear models with binary response, the logit Pearson residual of a success count is the loglinear one divided by $\sqrt{1-\hat{\pi}}$, hence larger.

Proof. Let the cell have count y of N trials. Fitted values agree, $\hat{\mu} = N\hat{\pi}$ (Theorem 8.16); the residuals are $(y - N\hat{\pi})/\sqrt{N\hat{\pi}(1-\hat{\pi})}$ and $(y - N\hat{\pi})/\sqrt{N\hat{\pi}}$, with ratio $1/\sqrt{1-\hat{\pi}} > 1$. ■

Standardized, the two residuals coincide, another reason to prefer them.

9.4 Modeling ordinal associations

Motivation. Nominal models ignore order; for ordinal variables a single trend parameter captures departures concentrated in the corners.

Example 9.9 (Premarital sex and teenage birth control)

1991 GSS. Rows: premarital sex always wrong (1), almost always wrong (2), wrong only sometimes (3), not wrong at all (4). Columns: teenage birth control available, strongly disagree (1) to strongly agree (4). Entries: count, (independence fit), [standardized residual], {L×L fit}.

	SD	D	A	SA
1	81 (42.4) [7.6] {80.9}	68 (51.2) [3.1] {67.6}	60 (86.4) [−4.1] {69.4}	38 (67.0) [−4.8] {29.1}
2	24 (16.0) [2.3] {20.8}	26 (19.3) [1.8] {23.1}	29 (32.5) [−0.8] {31.5}	14 (25.2) [−2.8] {17.6}
3	18 (30.1) [−2.7] {24.4}	41 (36.3) [1.0] {36.1}	74 (61.2) [2.2] {65.7}	42 (47.4) [−1.0] {48.8}
4	36 (70.6) [−6.1] {33.0}	57 (85.2) [−4.6] {65.1}	161 (143.8) [2.4] {157.4}	157 (111.4) [6.8] {155.5}

Independence fits poorly, $G^2(I) = 127.6$, df $= 9$; the corner residuals show a positive trend.

9.4.1 Linear-by-linear association in two-way tables

Definition (Linear-by-linear association (L×L) model)

With ordered row scores $u_1 \le \cdots \le u_I$ and column scores $v_1 \le \cdots \le v_J$,

$$\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y + \beta u_i v_j, \tag{9.5}$$

the special case $\lambda_{ij}^{XY} = \beta u_i v_j$ of the saturated model: one association parameter instead of $(I-1)(J-1)$. Independence is $\beta = 0$. With $u_i = i$, $v_j = j$ it is the *uniform association* model.

Theorem 9.10 (Odds ratios under L×L)

Under (9.5), for rows $a < c$ and columns $b < d$,

$$\log \frac{\mu_{ab}\mu_{cd}}{\mu_{ad}\mu_{cb}} = \beta(u_c - u_a)(v_d - v_b). \tag{9.6}$$

With unit-spaced scores all local odds ratios are e^{β}. If $\beta > 0$, Y tends to increase with X and μ_{ij} exceeds independence where X, Y are both high or both low.

Proof.

Step 1. By Theorem 8.3(b) the log odds ratio equals $\lambda_{ab}^{XY} + \lambda_{cd}^{XY} - \lambda_{ad}^{XY} - \lambda_{cb}^{XY}$ with $\lambda_{ij}^{XY} = \beta u_i v_j$:

$$\beta(u_a v_b + u_c v_d - u_a v_d - u_c v_b) = \beta(u_c - u_a)(v_d - v_b).$$

Step 2. For adjacent rows and columns with unit spacing, $(u_{i+1} - u_i)(v_{j+1} - v_j) = 1$.

Step 3. For $\beta > 0$ all odds ratios with $a < c$, $b < d$ exceed 1 (likelihood-ratio dependence); $|\beta u_i v_j|$ is largest at the corners. ■

$\beta u_i v_j$ is linear in the column scores within a row (slope βu_i) and in the row scores within a column. Scores are often midpoints or standardized (mean 0, variance 1 under the margins); then β is a log odds ratio per SD, and for an underlying bivariate normal $\beta \approx \rho/(1-\rho^2)$ ($\approx \rho$ when weak).

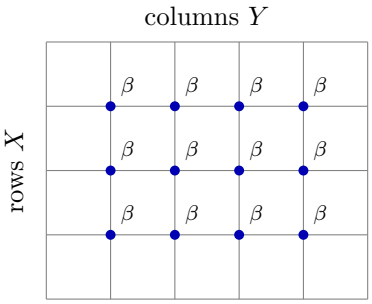


Figure 2: Uniform association: every 2×2 table of adjacent rows and columns has log odds ratio β (book Figure 9.2).

9.4.2 Corresponding logit model for adjacent responses

Proposition 9.11 (Adjacent-categories logit form)

Let $\pi_{j|i} = P(Y = j \mid X = i)$. Under (9.5),

$$\log \frac{\pi_{j+1|i}}{\pi_{j|i}} = (\lambda^Y_{j+1} - \lambda^Y_j) + \beta(v_{j+1} - v_j)u_i,$$

and for unit-spaced $\{v_j\}$ this is $\alpha_j + \beta u_i$: the same linear effect of X on all $J-1$ adjacent-categories logits.

Proof. $\pi_{j+1|i}/\pi_{j|i} = \mu_{i,j+1}/\mu_{ij}$; subtracting (9.5) at j from (9.5) at $j+1$, λ and λ^X_i cancel, leaving the stated expression. ■

9.4.3 Likelihood equations and model fitting

Theorem 9.12 (Likelihood equations for L×L)

The ML fit of (9.5) satisfies

$$\hat\mu_{i+} = n_{i+}, \qquad \hat\mu_{+j} = n_{+j}, \qquad \sum_{i,j} u_i v_j \hat\mu_{ij} = \sum_{i,j} u_i v_j n_{ij}.$$

So fitted and observed distributions share the score means, variances and correlation. Residual df $IJ - I - J$.

Proof.

Step 1. The Poisson log likelihood is

$$L = n\lambda + \sum_i n_{i+}\lambda^X_i + \sum_j n_{+j}\lambda^Y_j + \beta \sum_{i,j} u_i v_j n_{ij} - \sum_{i,j} \exp(\lambda + \lambda^X_i + \lambda^Y_j + \beta u_i v_j).$$

Setting derivatives in λ^X_i, λ^Y_j, β to zero gives the three sets of equations (sufficient statistics: the margins and $\sum u_i v_j n_{ij}$).

Step 2. Dividing by n, the fitted proportions $\hat\pi_{ij}$ and the sample proportions p_{ij} have equal margins, hence equal means and variances of u and of v, and equal $\sum u_i v_j \hat\pi_{ij} = \sum u_i v_j p_{ij}$, hence equal covariance and correlation.

Step 3. The model has $1 + (I-1) + (J-1) + 1$ parameters, so df $= IJ - (I + J)$, unsaturated except for 2×2 tables. ■

The fitted counts therefore reproduce the trend of the data. Fitting uses Newton–Raphson.

9.4.4 Sex opinions example

Example 9.13 (L×L fit to Example 9.9)

Scores $1, 2, 3, 4$: $G^2(\text{L} \times \text{L}) = 11.5$, df $= 8$, versus 127.6 on 9 df, the corners improving most. $\hat\beta = 0.286$ (SE 0.028): local odds ratio 1.33, CI $\exp(0.286 \pm 1.96 \times 0.028) = (1.26, 1.41)$. For the four corner cells, by (9.6),

$$\exp[0.286(4-1)(4-1)] = 13.1 = \frac{80.9 \times 155.5}{29.1 \times 33.0}.$$

Rescaled equal spacing (rows $2, 4, 6, 8$) rescales $\hat{\beta}$ (0.143), same fit. Scores $1, 2, 4, 5$: $G^2 = 8.8$, $\hat{\beta} = 0.146$. Standardized scores: $\hat{\beta} = 0.374$, so $\hat{\rho} = 0.333$ from $\beta = \rho/(1-\rho^2)$.

9.4.5 Directed ordinal test of independence

Within L×L, independence is $H_0 : \beta = 0$, tested by

$$G^2(I \mid \mathrm{L} \times \mathrm{L}) = G^2(I) - G^2(\mathrm{L} \times \mathrm{L}), \qquad \mathrm{df} = 1.$$

For Example 9.9: $127.6 - 11.5 = 116.1$; the Wald statistic $(\hat{\beta}/\mathrm{SE})^2 = 102.5$; the score statistic is the correlation statistic $M^2 = (n-1)r^2 = 112.6$ of Section 3.4.1.

Why the ordinal test is more powerful

Under L×L, $G^2(I \mid \mathrm{L} \times \mathrm{L})$ and $G^2(I)$ have about the same noncentrality but df 1 versus $(I-1)(J-1)$; power falls with df at fixed noncentrality, so the directed test is more powerful, increasingly so with I, J.

9.5 Poisson regression for rates

Motivation. For counts over unequal exposures, an offset models the rate while keeping the Poisson count as the observation.

9.5.1 Analyzing rates using loglinear models with offsets

Proposition 9.14 (Rate model with offset)

For counts n_i with exposure (index) t_i, the model

$$\log(\mu_i/t_i) = \alpha + \beta x_i \tag{9.7}$$

is the Poisson model $\log \mu_i = \log t_i + \alpha + \beta x_i$ with offset $\log t_i$ (coefficient fixed at 1): $\mu_i = t_i \exp(\alpha + \beta x_i)$, and e^{β} is the multiplicative effect on the rate.

Proof. Add $\log t_i$ to both sides of (9.7) and exponentiate. The ratio of rates at $x+1$ and x is $e^{\alpha+\beta(x+1)}/e^{\alpha+\beta x} = e^{\beta}$. ■

The identity link $\mu_i = \alpha t_i + \beta x_i t_i$ needs no offset (predictors t_i, $x_i t_i$, no intercept); effects are additive, and fitting may fail with negative fitted values.

9.5.2 Modeling death rates for heart valve operations

Example 9.15 (Heart valve replacement)

109 patients by valve type and age; time at risk = total follow-up (months) in the cell, deaths and censored alike.

Age		Aortic	Mitral
< 55	Deaths	4	1
	Time at risk	1259	2082
	Death rate	0.0032	0.0005
55+	Deaths	7	9
	Time at risk	1417	1647
	Death rate	0.0049	0.0055

A binary model (died yes/no) would ignore the unequal exposures (3 to 97 months). With dummies a (age 55+) and v (mitral),

$$\log(\mu_{ij}/t_{ij}) = \alpha + \beta_1 a_i + \beta_2 v_j. \tag{9.8}$$

$\hat{\beta}_1 = 1.221$ (SE 0.514), $\hat{\beta}_2 = -0.330$ (SE 0.438). Given valve type the older death rate is $e^{1.221} = 3.4$ times the younger; Wald CI $\exp(1.221 \pm 1.96 \times 0.514) = (1.2, 9.3)$, LR CI $(1.3, 10.4)$. $G^2 = 3.2$, $X^2 = 3.1$, df $= 1$; without valve type $G^2 = 3.8$ (df 2); without age the fit is poor. Fitted rates: < 55: 0.0018 (aortic), 0.0013 (mitral); 55+: 0.0062, 0.0044. Identity link: $G^2 = 1.1$ (df 1), $\hat{\beta}_1 = 0.0040$ (SE 0.0014), an additive rate difference. Only 21 of 109 died, so estimates are imprecise, but all patients contribute through their times at risk.